Jacob Blazina CS 350 Task 2: Project Grounding and Conceptualization [Release Version]

A.

1. acceleration: the measure of how fast an aircraft is speeding up or slowing down
   - i source: [@physics.info/acceleration/]
   - ii category: **noun**
   - iii data: change in velocity over time (**dynamic** because the acceleration can constantly change)
   - iv control: speeds up the plane or slows down the plane
   - v behavior: the engines current change in output determines acceleration
   - vi role: **output** because acceleration is the effect that is caused by the engine
   - vii pattern: **behavioral** because it is affected by other aspects of the machine
   - viii concern: **view** because it represents the current state of the plane in context to it's other parameters
   - ix layer: **ME** because it is part of the physical representation of the plane in relation to the world
   - x difficulty: **hard** because many other variable factors contribute to the acceleration
   - xi risk: **moderate** because acceleration could affect flight and movement simulations but we could still try out other things without knowing the acceleration
   - xii confidence: **high** because I enjoy physics
   - xiii presentation: text numbers representing distance / time^2
2. actuator: a mechanical device that translates one type of motion to another type of motion
   - i source: [@www.progressiveautomations.com/blogs/products/linear-actuators-airplanes]
   - ii category: **noun**
   - iii data: an electric or hydraulic servo couple with at least 2 mounting points (**dynamic** because it changes state)
   - iv control: can extend, retract, or rotate to cause another part of the aircraft to move in a certain way
   - v behavior: an electrical signal gets supplied to a mechanical device to induce physical motion
   - vi role: **processing** because it receives an electrical input and converts it to a mechanical output
   - vii pattern: **structural** because it bridges the electrical and mechanical systems of the airplane
   - viii concern: **controller** because it facilitates input from the model into state for the view
   - ix layer: **ME** because it facilitates physical movement
   - x difficulty: **easy** because we take a simple input and produce a simple output
   - xi risk: **high** because if our flaps or landing gear don't work the aircraft won't work
   - xii confidence: **moderate** because there are many different types of actuators on an aircraft
   - xiii presentation: two squiggly intersecting arrows
3. aileron: a small hinged flap on the outermost part of the airplane wing, used to generate rolling motion in an aircraft
   - i source: [@www.grc.nasa.gov/www/k-12/airplane/alr.html]
   - ii category: **noun**
   - iii data: a flap on the rearward-facing outer portion of the wing that pivots up or down to cause a rolling motion. (**dynamic** because the flaps will change position often)
   - iv control: the flaps pivot up or down on either wing
   - v behavior: an electrical signal to an actuator will trigger a change in position of the ailerons to modify the roll angle of the aircraft
   - vi role: **output** because it is a mechanical device that changes position based on input from the operator/computer
   - vii pattern: **behavioral** because it relies on input and changes the position of the aircraft
   - viii concern: **model** because the position of the flaps effects the motion of the plane
   - ix layer: **ME** because it is a physical element that interacts with the physical element of air movement
   - x difficulty: **moderate** because we have a simple input that needs to correspond with the inverse movements of multiple flaps
   - xi risk: **moderate** because the flaps are critical for correct operation of the aircraft, but since our plane is not flying it won't affect anything
   - xii confidence: **high** because NASA has a really good demonstration
   - xiii presentation: two arrows drawn in a circle with the tails at the top and points at the bottom
4. airspeed: the relative velocity between the aircraft and the air, computed by the vector difference of the ground speed and the wind speed
   - i source: [@www.grc.nasa.gov/www/k-12/airplane/move.html]
   - ii category: **noun**

iii data: the speed of the aircraft calculated by the relationship of ground speed and wind speed. (dynamic because the airspeed is always changing, even if the aircraft isn't moving the wind can affect airspeed)
iv control: the velocity of the wind and the velocity of the aircraft affect airspeed
v behavior: the airspeed is the most important factor regarding if an aircraft can achieve lift-off or landing using a certain length of runway.
vi Role: **output** because wind speed and ground speed due to thrust get processed into airspeed
vii Pattern: **behavioral** because the airspeed has a direct effect on the movement of the aircraft
viii Concern: **view** because the airspeed represents the state of the aircraft
ix Layer: **ME** because speed relates to the aircrafts relationship with its surroundings
x Difficulty: **moderate** because we can calculate airspeed using straight forward vector mathematics
xi Risk: **moderate** because airspeed is the critical factor for flight, but since our aircraft is not flying it won't affect that
xii Confidence: **moderate** because airspeed is a combination of external and internal forces
xiii Presentation: a speedometer

5. Altitude: the height of the aircraft in relation to sea level
   i Source: [@www.grc.nasa.gov/www/k-12/airplane/density.html]
   ii Category: **noun**
   iii Data: measured in feet or meters, denotes the total distance an aircraft is located above the sea level line (**dynamic** because it changes)
   iv Control: as altitude increases, air density decreases
   v Behavior: as air density decreases to a certain point, aircraft can no longer maintain flight
   vi Role: **input** because altitude changes even at ground level
   vii Pattern: **behavioral** because altitude has a direct effect on the movement of the aircraft
   viii Concern: **model** because altitude starts as a fixed value and gets modified by the aircraft and affects many things in the aircraft
   ix Layer: **ME** because it affects aerodynamic properties
   x Difficulty: **easy** because it is a simple number and is easy to calculate
   xi Risk: **moderate** because altitude is a critical factor for flight, but since our aircraft is not flying it won't affect it too much
   xii Confidence: **moderate** because altitude is a combination of the state of the aircraft and can also be controlled by the aircraft
   xiii Presentation: horizontal lines stacked in vertical order
6. angle-of-attack indicator: the sensor that indicates the angle between the aircraft and the oncoming air
   i source: [@www.flyingmag.com/how-it-works-angle-attack-indicator]
   ii category: **noun**
   iii data: a probe, computer unit, and indicator (**static** because it stays put)
   iv control: passes the information to the operator via sight and/or sound, sends data to computer
   v behavior: air passes through a hole on the longitudinal axis and another one 45 degrees downward and compares the incoming air pressure to the static air pressure
   vi role: **processing** because it takes input from the aircraft's relationship to the air and feeds it to the operator
   vii pattern: **structural** because it communicates external data to the operator
   viii concern: **view** because it indicates the state of the aircraft in relation to the outside world
   ix layer: **EE** because it translates air information into electrical signals
   x difficulty: **moderate** because it can be easy to compare two vector angles but hard to calculate air pressure and air density
   xi risk: **moderate** because it is essential for the flight of the aircraft but since our aircraft is not flying it will always give us the same reading
   xii confidence: **moderate** because the AoA indicator is very specific for each aircraft and can perform several functions
   xiii presentation: two arrows connected at the tail forming an angle from 0 degrees to 45 degrees
7. autopilot: system that controls some flight functions
   i source: [@www.rd.com/advice/travel/how-airplane-autopilot-works/]
   ii category: **noun**
   iii data: computer system and controls (**dynamic** because it has different modes)
   iv control: can hold heading, navigation, altitude, and approaches
   v behavior: system uses sensors, GPS, and other subsystems to automate processes

vi role: **processing** because it takes inputs from subsystems and outputs things to control the aircraft (**dynamic**)
vii pattern: **behavioral** because it is constantly doing something to fly the aircraft
viii concern: **controller** because it is facilitating the relationship between the models and the views
ix layer: **CS** because it is a computer system
x difficulty: **Hard** because there are many inputs and many outputs to deal with
xi risk: **low** because the aircraft can be operated without it
xii confidence: **moderate** because it is a very complicated system that varies from aircraft to aircraft
xiii presentation: an aircraft symbol with an A next to it

8. autothrottle: automatically adjusts speed or angle
    i source: [@ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/19880010924.pdf]
    ii category: **noun**
    iii data: computer system (**dynamic** because it has different modes)
    iv control: controls flightpath angle and velocity
    v behavior: uses elevator and thrust to control angle and speed in a common fashion
    vi role: **processing** because it takes inputs from subsystems and sends output information to other subsystems
    vii pattern: **behavioral** because it is constantly working to control the aircraft
    viii concern: **controller** because it is facilitating the relationship between the models and views
    ix layer: **CS** because it is a computer system
    x difficulty: **Hard** because it has to differentiate between maneuvers requiring a net energy change
    xi risk: **moderate** because the aircraft can be operated without it but would be difficult to do so
    xii confidence: **low** because there are many different autothrottle systems
    xiii presentation: a speedometer with an A next to it
9. communication bus: serial communications hardware
    i source: [@www.aviationtoday.com/2009/05/01/can-bus-in-aviation/]
    ii category: **noun**
    iii data: LRUs and bus (**static** because it is a physical controller)
    iv control: transmits and receives data
    v behavior: uses a protocol to facilitate communication between subsystems
    vi role: **processing** because it is responsible for moving information around
    vii pattern: **structural** because it facilitates communication between separate subsystems
    viii concern: **controller** because it takes information from models and views and lets them talk to each other
    ix layer: **EE** because it is the physical device that deals with the electrical signals
    x difficulty: **Hard** because it is a complicated piece of hardware that facilitates many-to-many relationships
    xi risk: **high** because if subsystems can't talk to each other nothing will work
    xii confidence: **high** because a communication bus is common in a lot of modern machinery
    xiii presentation: a small box with a grid
10. communication bus protocol: the system that the bus hardware uses to facilitate the communication
    i source: [@www.aviationtoday.com/2009/05/01/can-bus-in-aviation/]
    ii category: **noun**
    iii data: data encoding, transmission, filtering (**static** because the protocol stays the same)
    iv control: sends signals to the common bus and filters it for each line attached that get's it own message
    v behavior: allows one signal to facilitate communication for multiple lines without cross contamination
    vi role: **processing** because it is responsible for facilitating communication between the bus hardware and the LRUs
    vii pattern: **structural** because it facilitates communication between the bus hardware and the subsystems
    viii concern: **controller** because it moves information around between subsystems
    ix layer: **EE** because signals are sent and filtered at the lowest level: bits
    x difficulty: **moderate** because you want one information stream but multiple systems feeding off it
    xi risk: **high** because subsystems need to talk to each other
    xii confidence: **moderate** because its hard to distinguish between what the bus does and what the protocol does
    xiii presentation: a small box with lines in it
11. elevator: the hinged part of the horizontal stabilizer
    i source: [@www.grc.nasa.gov/www/k-12/airplane/airplane.html]
    ii category: **noun**
    iii data: hinge and a flap (**dynamic** because the flap moves)

    iv control: receives signal from piloting system
    v behavior: move upwards or downwards to adjust the pitch of the aircraft
    vi role: **output** because it receives a command and moves up or down
    vii pattern: **behavioral** because it positions the plane in 3d space
    viii concern: **view** because it represents the positioning of the plane in 3d space
    ix layer: **ME** because it is a physical moving part
    x difficulty: **easy** because it is trivial to signal a flap to move up or down
    xi risk: **high** because it supplies the downforce needed on the tail
    xii confidence: **high** because it is a simple, physical part
    xiii presentation: a construction level

12. engine: provides the thrust needed to propel an aircraft forward
    i source: [@www.grc.nasa.gov/www/k-12/airplane/turbine.html]
    ii category: **noun**
    iii data: turbine nozzle burner compressor inlet (**dynamic** because the engine has different modes)
    iv control: pressurizes incoming air and mixes with fuel to create thrust
    v behavior: the thrust from the engines propel the aircraft forward
    vi role: **output** because engines receive instruction from the pilot system and creates thrust
    vii pattern: **behavioral** because it causes the aircraft to move
    viii concern: **controller** because it moves the aircraft around in space
    ix layer: **ME** because it is a physical object
    x difficulty: **moderate** because it is easy to program the thrust but hard to program the specifics of the engine
    xi risk: **high** because no engines mean no aircraft
    xii confidence: **moderate** because there are many types of aircraft engine
    xiii presentation: check engine light symbol

13. FADEC: Full Authority Distributed Engine Control
    i source: [@ntrs.nasa.gov/archive/nasa/casi.ntrs.nasa.gov/20130013439.pdf]
    ii category: **noun**
    iii data: computer-based controller (**dynamic** because it has different modes)
    iv control: provides the command signals that get sent to the engine
    v behavior: a pilot can make throttle adjustments but the final command is always determined by the computer
    vi role: **processing** because it takes inputs and issues commands to the engine
    vii pattern: **structural** because it is the link between the operators and the engines
    viii concern: **controller** because it causes the engine to perform as needed
    ix layer: **CS** because there are programs that determine the behavior
    x difficulty: **hard** because it is a very complicated computer system that needs much testing, as it takes control away from the human
    xi risk: **high**, if it malfunctions then the human cannot bypass it
    xii confidence: **low** because it is a very complicated system that relies on many subsystems
    xiii presentation: computer screen

14. flight control law: control modes of flights
    i source: [@www.skybrary.aero/index.php/Flight_Control_Laws]
    ii category: **noun**
    iii data: computational laws which assign flight control modes (**static** because they are strongly defined)
    iv control: computer determines the operational state
    v behavior: flight modes determine how the aircraft operates whilst in the air
    vi role: **input** because the current mode determines how the aircraft is going to operate
    vii pattern: **creational** because it sets the flight information for the whole aircraft
    viii concern: **model** because it holds aircraft information
    ix layer: **CS** because it is a software level thing
    x difficulty: **hard** because it takes inputs and has to decide on the current flight mode
    xi risk: **moderate** because it needs to know when we are on the ground so it doesn't try to fly
    xii confidence: **low** because it could be more software or hardware related
    xiii presentation: squiggly lines

15. flap

i source: [@www.grc.nasa.gov/www/k-12/airplane/flap.html]
ii category: **noun**
iii data: the physical component that moves to control the airplane in 3d space (**dynamic** because the flaps move)
iv control: moves up or down
v behavior: flight system commands the flap to move to adjust the position of the airplane
vi role: **output** because the flight system commands the flaps to move
vii pattern: **behavioral** because the flaps move
viii concern: **view** because we need to know the position of the flaps
ix layer: **ME** because it is a physical device
x difficulty: **easy** because we just need to set it's position
xi risk: **high** because the flaps are pertinent to testing out the airplane even on the ground
xii confidence: **high** because they are a simple, important part
xiii presentation: a wing outline

16. flight control system: the means by which a pilot controls the direction and attitude in flight
    i source: [@www.skybrary.aero/index.php/Flight_Controls]
    ii category: **noun**
    iii data: ailerons, elevators and rudder (**dynamic** because of many moving parts)
    iv control: flight control surfaces move to change direction in flight
    v behavior: input from the operator gets sent to the flight control surfaces
    vi role: **output** because they are all physical parts
    vii pattern: **behavioral** because it moves the plane
    viii concern: **view** because they change
    ix layer: **ME** because it is a physical device
    x difficulty: **moderate** because it the combination of multiple parts
    xi risk: **high** because it's what we are testing on the aircraft
    xii confidence: **moderate** because its multiple parts
    xiii presentation: joystick
17. flight data recorder: records aircraft information
    i source: [@www.skybrary.aero/index.php/Flight_Data_Recorder_(FDR)]
    ii category: **noun**
    iii data: flight recorder hardware (**static** because it does the same thing all the time)
    iv control: records information about the aircraft as well as voice records the cockpit
    v behavior: stores information to be viewed later
    vi role: **input** because it records things
    vii pattern: **creational** because it creates recordings
    viii concern: **model** because it holds information
    ix layer: **CS** because it is a computer system
    x difficulty: **hard** because it saves data from the whole aircraft
    xi risk: **low** because our plane won't crash
    xii confidence: **moderate** because there are various types of recorders
    xiii presentation: microphone
18. fly-by-wire control system: a flight control system using computers and not direct operator input
    i source: [@www.skybrary.aero/index.php/Fly-By-Wire]
    ii category: **noun**
    iii data: computers and electrical signals (**static** because it does the same thing all the time)
    iv control: computer controls airplane
    v behavior: operator input sends signals to computer that converts into aircraft control
    vi role: **processing** because it processes operator actions into aircraft control
    vii pattern: **structural** because it connects operator to aircraft control
    viii concern: **controller** because it connects operator to aircraft control
    ix layer: **EE** because it is primarily electrical signals
    x difficulty: **hard** because it is the interface between the operator and the aircraft
    xi risk: **high** because it is important to control our aircraft
    xii confidence: **low** because it is a mix of ME, EE and CS

xiii presentation: two dots with a line

19. horizontal stabilizer: the small wing on the tail of the aircraft
    i source: [@www.grc.nasa.gov/www/k-12/airplane/elv.html]
    ii category: **noun**
    iii data: fixed wing section and elevators (**static** because the stabilizer doesn't move)
    iv control: helps control pitch
    v behavior: the stabilizer doesn't move but the elevators attached to it do
    vi role: **output** because the position is determined by the aircraft
    vii pattern: **behavioral** because it helps control the plane
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it is a physical thing
    x difficulty: **easy** because it is a static constant
    xi risk: **moderate** because it wouldn't be a proper plane
    xii confidence: **high** because it is a straight-forward part
    xiii presentation: horizontal double sided arrow
20. latitude/longitude: global coordinate system
    i source: [@www.skybrary.aero/index.php/Inertial_Navigation_System_(INS)]
    ii category: **noun**
    iii data: relation of an object to the world (**static** because the system never changes)
    iv control: degree and minutes designation
    v behavior: changes depending on position on the planet
    vi role: **output** because the position of the plane determines it
    vii pattern: **behavioral** because it changes all the time
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it's a physical thing
    x difficulty: **easy** because it's a 2d coordinate system
    xi risk: **low** because we aren't going anywhere
    xii confidence: **high** because it's a common factor in the world
    xiii presentation: compass rose
21. MCAS: maneuvering characteristics augmentation system
    i source: [@www.boeing.com/commercial/737max/737-max-software-updates.page]
    ii category: **noun**
    iii data: software system (**dynamic** because it has different modes)
    iv control: receives sensor input and adjust plane movement
    v behavior: moves the plane's nose up or down depending on angle of attack sensors
    vi role: **processing** because it takes sensor data and turns it movement output
    vii pattern: **behavioral** because it controls movement
    viii concern: **controller** because it controls movement
    ix layer: **CS** because its in the software
    x difficulty: **hard** because it is a complex program
    xi risk: **high** because crashing
    xii confidence: **moderate** because it's a combination of things
    xiii presentation: big M with a circle
22. pitch: a rotation axis of an airplane
    i source: [@www.grc.nasa.gov/www/k-12/airplane/rotations.html]
    ii category: **noun**
    iii data: up/down rotation angle (**dynamic** because the value changes)
    iv control: plane gaining altitude or decreasing altitude
    v behavior: nose down goes to the ground, nose up gains altitude
    vi role: **output** because the flight control surfaces dictate the pitch angle
    vii pattern: **behavioral** because the pitch changes
    viii concern: **view** because it is the state of the aircraft
    ix layer: **ME** because physical property
    x difficulty: **moderate** because it can be a number in the program

    - xi risk: **high** because it determines a lot of characteristics
    - xii confidence: **high** because it is a common measurement
    - xiii presentation: an angle

23. pitot tube: sensor used as a speedometer
    - i source: [@www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/pitot.html]
    - ii category: **noun**
    - iii data: a tube with holes (**static** because it stays in place and operates consistently)
    - iv control: air goes in and pressure gets measured
    - v behavior: the pressure difference between holes calculates air density and air speed
    - vi role: **input** because it is a sensor
    - vii pattern: **creational** because it provides values
    - viii concern: **view** because it represents the state
    - ix layer: **ME** because it is physical properties
    - x difficulty: **easy** because we don't calculate airpressures in our program
    - xi risk: **low** because we can calculate speed more simply
    - xii confidence: **moderate** because we could have to use more data to calculate speed
    - xiii presentation: speedometer
24. roll: an axis of rotation
    - i source: [@www.grc.nasa.gov/www/k-12/airplane/rotations.html]
    - ii category: **noun**
    - iii data: side to side rotation of the airplane (**dynamic** because the value changes)
    - iv control: which direction the plane is tilting
    - v behavior: causes the plane to turn
    - vi role: **output** because the flight control surfaces dictate the roll angle
    - vii pattern: **behavioral** because roll changes
    - viii concern: **view** because it represents the state of the plane
    - ix layer: **ME** because it's a physical property
    - x difficulty: **moderate** because it can be a number in the program
    - xi risk: **high** because it determines a lot about the plane
    - xii confidence: **high** because it is a common measurement
    - xiii presentation: circular arrows.
25. rudder: rotating part of the vertical stabilizer
    - i source: [@www.grc.nasa.gov/www/k-12/airplane/rud.html]
    - ii category: **noun**
    - iii data: rotating flap (**dynamic** because it moves)
    - iv control: moves side to side
    - v behavior: rudder changes the yaw
    - vi role: **output** because it gets controlled by the operator
    - vii pattern: **behavioral** because it causes the plane to move
    - viii concern: **view** because it represents the state of the plane
    - ix layer: **ME** because its a physical thing
    - x difficulty: **easy** because it's a simple position value
    - xi risk: **high** because it is important for the plane
    - xii confidence: **high** because it is a simple physical part
    - xiii presentation: up and down flap
26. sensor: physical device that sends external data to a system
    - i source: [@electronics360.globalspec.com/article/9934/what-you-need-to-know-about-aircraft-sensors]
    - ii category: **noun**
    - iii data: physical device and electrical signals (**static** because a sensor has one job)
    - iv control: sends information to a computer
    - v behavior: converts physical readings to electric signals
    - vi role: **input** because it extracts information from the outside world
    - vii pattern: **creational** because it gleans outside information
    - viii concern: **model** because it represents information

ix layer: **ME** because it is a physical device
x difficulty: **easy** because we get the information directly
xi risk: **high** because sensor readings are required for the craft to operate
xii confidence: **moderate** because there are many types of sensors
xiii presentation: diamond

27. slat: moving part on the leading edge of the wing
    i source: [@www.grc.nasa.gov/www/k-12/airplane/flap.html]
    ii category: **noun**
    iii data: metal piece that moves on tracks (**dynamic** because it moves)
    iv control: extends and retracts and rotates
    v behavior: extends to create more wing area and increase lift
    vi role: **output** because it moves to control the plane
    vii pattern: **behavioral** because it moves the plane
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it is a physical part
    x difficulty: **easy** because it has simple movements
    xi risk: **high** because it is critical for the function of the plane
    xii confidence: **high** because it is a primary attribute of an airplane wing
    xiii presentation: wing with an arrow pointing to the leading edge
28. speed brake
    i source: [@www.skybrary.aero/index.php/Spoilers_And_Speedbrakes]
    ii category: **noun**
    iii data: section that can stick out to increase drag and reduce speed (**dynamic** because it moves)
    iv control: extends and retracts
    v behavior: reduces speed
    vi role: **output** because it moves to control the plane
    vii pattern: **behavioral** because it moves the plane
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it's a physical part
    x difficulty: **easy** because we can set it to extend or retract
    xi risk: **low** because not all planes have them
    xii confidence: **low** because not all planes use them
    xiii presentation: stop sign
29. stall: a craft no longer producing lift when the angle attack angle is exceeded
    i source: [@www.skybrary.aero/index.php/Stall]
    ii category: **noun**
    iii data: the lack of lift (**dynamic** because it's a state)
    iv control: enters stall or exits stall state
    v behavior: a plane that stalls falls out of the sky
    vi role: **output** because it is determined by the state of the plane
    vii pattern: **behavioral** because it represents the state of the plane
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it's a physical property
    x difficulty: **easy** because we can calculate it using angle of attack
    xi risk: **low** because our plane won't be airborne
    xii confidence: **moderate** because other factors affect stall
    xiii presentation: big X
30. static port: sensor to detect static pressure
    i source: [@www.skybrary.aero/index.php/Pitot_Static_System]
    ii category: **noun**
    iii data: sensor hardware (**static** because its called a static port)
    iv control: outputs static air pressure
    v behavior: pressure is measured through vents that are in neutral locations on the plane
    vi role: **input** because it is a sensor reading

vii pattern: **creational** because it is gleaning outside information
viii concern: **model** because it represents external data
ix layer: **ME** because it's a physical sensor
x difficulty: **easy** because it's just a number in our program
xi risk: **low** because our static pressure won't change
xii confidence: **moderate** because the sensors can be complex but the program may be simple
xiii presentation: square

31. trim: an adjustment that maintains a set attitude
    i source: [@www.skybrary.aero/index.php/Trim_Systems]
    ii category: **noun**
    iii data: a tab and an adjuster (**dynamic** because trim changes)
    iv control: small adjustment that causes aerodynamic deflection
    v behavior: can be manually or automatically adjusted to maintain appropriate movement
    vi role: **input** because it affects movement of the plane
    vii pattern: **behavioral** because it affects the movement of the plane
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it is a physical thing
    x difficulty: **moderate** because it is a minute adjustment based on current operating data
    xi risk: **low** because our plane is not flying
    xii confidence: **low** because there are 3 types of trim systems
    xiii presentation: calipers
32. velocity: the speed and direction of an aircraft
    i source: [@www.grc.nasa.gov/www/k-12/airplane/vel.html]
    ii category: **noun**
    iii data: change in position over time (**dynamic** because velocity changes)
    iv control: increase or decrease in velocity
    v behavior: causes the aircraft to move a certain distance in a certain time
    vi role: **output** because it's determined by the aircraft
    vii pattern: **behavioral** because it changes
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it's a physical thing
    x difficulty: **easy** because it's a number
    xi risk: **low** because the plane is not going anywhere
    xii confidence: **high** because it's physics
    xiii presentation: MPH
33. yaw: axis of rotation

    i source: [@www.grc.nasa.gov/www/k-12/airplane/yaw.html]
    ii category: **noun**
    iii data: the angle of side to side motion of an aircraft (**dynamic** because it's a state)
    iv control: rotates left or right
    v behavior: points the nose of the airplane
    vi role: **output** because it's determined by the aircraft
    vii pattern: **behavioral** because it changes
    viii concern: **view** because it represents the state of the plane
    ix layer: **ME** because it's a physical thing
    x difficulty: **easy** because it's a number
    xi risk: **low** because the plane is not going anywhere
    xii confidence: **high** because it's a common measurement
    xiii presentation: triangle

B.

Flight Control Surface	Physics	Operational Hardware	Operational Software
Aileron	Acceleration	actuator	autopilot
Elevator	Airspeed	angle-of-attack indicator	autothrottle
Engine	Altitude	communication bus	communication bus protocol
Flap	Latitude/longitude	Flight data recorder	FADEC
Horizontal stabilizer	Pitch	Pitot tube	Flight control law
Rudder	Roll	Sensor	MCAS
Slat	Stall	Static port	Flight control system
Speed brake	Velocity		
trim	yaw		

Total Word Count: 4724